

```
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestClassifier

df=pd.read_csv("/content/drive/MyDrive/summa/Book1.csv",sep=",")
print(df.head())
```

	State	Literacy	Cleanliess	Crime_Rate	Output
0	A	55	88	44	1
1	B	44	56	33	1
2	C	24	33	55	0
3	D	15	22	88	1
4	E	24	33	29	0

```
df.isnull().sum()
```

```
State      0
Literacy   0
Cleanliess 0
Crime_Rate 0
Output     0
dtype: int64
```

```
feature=df[['Literacy','Cleanliess','Crime_Rate']]
target=df['Output']
```

```
RF=RandomForestClassifier()
RF.fit(feature,target)
```

```
RandomForestClassifier()
```

```
RandomForestClassifier(criterion='entropy')
```

```
RandomForestClassifier(criterion='entropy')
```

```
RF.score(feature,target)
```

```
1.0
```

```
l=int(input("enter literacy: "))
cl=int(input("enter cleanliness: "))
cr=int(input("enter crime rate: "))
pred=RF5.predict([[l,cl,cr]])
if pred==1:
    print("Good State")
else:
    print("Bad State")
```

```
enter literacy: 43
enter cleanliness: 48
enter crime rate: 23
Good State
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/base.py:493:
UserWarning: X does not have valid feature names, but
RandomForestClassifier was fitted with feature names
  warnings.warn(
```

```
RF.estimators_
```

```
[DecisionTreeClassifier(max_features='sqrt', random_state=410709367),
 DecisionTreeClassifier(max_features='sqrt', random_state=1679972287),
 DecisionTreeClassifier(max_features='sqrt', random_state=675758136),
 DecisionTreeClassifier(max_features='sqrt', random_state=113731617),
 DecisionTreeClassifier(max_features='sqrt', random_state=867635798),
 DecisionTreeClassifier(max_features='sqrt', random_state=1622166036),
 DecisionTreeClassifier(max_features='sqrt', random_state=378271317),
 DecisionTreeClassifier(max_features='sqrt', random_state=76239352),
 DecisionTreeClassifier(max_features='sqrt', random_state=1448431251),
 DecisionTreeClassifier(max_features='sqrt', random_state=226943305),
 DecisionTreeClassifier(max_features='sqrt', random_state=290510170),
 DecisionTreeClassifier(max_features='sqrt', random_state=817632727),
 DecisionTreeClassifier(max_features='sqrt', random_state=1088511639),
 DecisionTreeClassifier(max_features='sqrt', random_state=1238626830),
 DecisionTreeClassifier(max_features='sqrt', random_state=1077220788),
 DecisionTreeClassifier(max_features='sqrt', random_state=1607288888),
 DecisionTreeClassifier(max_features='sqrt', random_state=84818130),
 DecisionTreeClassifier(max_features='sqrt', random_state=531913329),
 DecisionTreeClassifier(max_features='sqrt', random_state=916118369),
 DecisionTreeClassifier(max_features='sqrt', random_state=296103654),
 DecisionTreeClassifier(max_features='sqrt', random_state=1882245818),
 DecisionTreeClassifier(max_features='sqrt', random_state=636717461),
 DecisionTreeClassifier(max_features='sqrt', random_state=454774408),
 DecisionTreeClassifier(max_features='sqrt', random_state=1758475640),
 DecisionTreeClassifier(max_features='sqrt', random_state=443147776),
 DecisionTreeClassifier(max_features='sqrt', random_state=1066075558),
 DecisionTreeClassifier(max_features='sqrt', random_state=944117543),
 DecisionTreeClassifier(max_features='sqrt', random_state=706817074),
 DecisionTreeClassifier(max_features='sqrt', random_state=93918721),
 DecisionTreeClassifier(max_features='sqrt', random_state=1649107757),
 DecisionTreeClassifier(max_features='sqrt', random_state=1446632205),
 DecisionTreeClassifier(max_features='sqrt', random_state=1406398323),
 DecisionTreeClassifier(max_features='sqrt', random_state=1913049748),
 DecisionTreeClassifier(max_features='sqrt', random_state=427867945),
 DecisionTreeClassifier(max_features='sqrt', random_state=1695055763),
 DecisionTreeClassifier(max_features='sqrt', random_state=1153739254),
 DecisionTreeClassifier(max_features='sqrt', random_state=1517740577),
 DecisionTreeClassifier(max_features='sqrt', random_state=137407910),
 DecisionTreeClassifier(max_features='sqrt', random_state=1130916328),
 DecisionTreeClassifier(max_features='sqrt', random_state=1035208898),
 DecisionTreeClassifier(max_features='sqrt', random_state=1537375203),
 DecisionTreeClassifier(max_features='sqrt', random_state=1529725335),
 DecisionTreeClassifier(max_features='sqrt', random_state=1857976015),
```

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DecisionTreeClassifier(max_features='sqrt', random_state=1357676754),
DecisionTreeClassifier(max_features='sqrt', random_state=1885207470),
DecisionTreeClassifier(max_features='sqrt', random_state=1610337076),
DecisionTreeClassifier(max_features='sqrt', random_state=732937633),
DecisionTreeClassifier(max_features='sqrt', random_state=642244079),
DecisionTreeClassifier(max_features='sqrt', random_state=1613542489),
DecisionTreeClassifier(max_features='sqrt', random_state=1412291467),
DecisionTreeClassifier(max_features='sqrt', random_state=342959646),
DecisionTreeClassifier(max_features='sqrt', random_state=1895124904),
DecisionTreeClassifier(max_features='sqrt', random_state=1018623378),
DecisionTreeClassifier(max_features='sqrt', random_state=822262318),
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DecisionTreeClassifier(max_features='sqrt', random_state=1900246782),
DecisionTreeClassifier(max_features='sqrt', random_state=1002312352),
DecisionTreeClassifier(max_features='sqrt', random_state=1393702873),
DecisionTreeClassifier(max_features='sqrt', random_state=483936198),
DecisionTreeClassifier(max_features='sqrt', random_state=739976395),
DecisionTreeClassifier(max_features='sqrt', random_state=1582547578),
DecisionTreeClassifier(max_features='sqrt', random_state=1132060516),
DecisionTreeClassifier(max_features='sqrt', random_state=1020035141),
DecisionTreeClassifier(max_features='sqrt', random_state=2084707313),
DecisionTreeClassifier(max_features='sqrt', random_state=734217463),
DecisionTreeClassifier(max_features='sqrt', random_state=1363724590),
DecisionTreeClassifier(max_features='sqrt', random_state=450368875),
DecisionTreeClassifier(max_features='sqrt', random_state=638081250),
DecisionTreeClassifier(max_features='sqrt', random_state=2015050291),
DecisionTreeClassifier(max_features='sqrt', random_state=1125607462),
DecisionTreeClassifier(max_features='sqrt', random_state=947358123),
DecisionTreeClassifier(max_features='sqrt', random_state=1391281659),
DecisionTreeClassifier(max_features='sqrt', random_state=1415658672),
DecisionTreeClassifier(max_features='sqrt', random_state=1325273507),
DecisionTreeClassifier(max_features='sqrt', random_state=428469757),
DecisionTreeClassifier(max_features='sqrt', random_state=1026884100),
DecisionTreeClassifier(max_features='sqrt', random_state=1081520320),
DecisionTreeClassifier(max_features='sqrt', random_state=1023448176),
DecisionTreeClassifier(max_features='sqrt', random_state=1480473985),
DecisionTreeClassifier(max_features='sqrt', random_state=40266160),
DecisionTreeClassifier(max_features='sqrt', random_state=798178710),
DecisionTreeClassifier(max_features='sqrt', random_state=1944909932),
DecisionTreeClassifier(max_features='sqrt', random_state=677982107),
DecisionTreeClassifier(max_features='sqrt', random_state=640598509),
DecisionTreeClassifier(max_features='sqrt', random_state=2060124302),
DecisionTreeClassifier(max_features='sqrt', random_state=40279845),
DecisionTreeClassifier(max_features='sqrt', random_state=1927747276),
DecisionTreeClassifier(max_features='sqrt', random_state=2048923950),
DecisionTreeClassifier(max_features='sqrt', random_state=189417067),
DecisionTreeClassifier(max_features='sqrt', random_state=1055279356),
DecisionTreeClassifier(max_features='sqrt', random_state=1756872983),
DecisionTreeClassifier(max_features='sqrt', random_state=18754862),
```

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DecisionTreeClassifier(max_features='sqrt', random_state=1638803943),
DecisionTreeClassifier(max_features='sqrt', random_state=492299998),
DecisionTreeClassifier(max_features='sqrt', random_state=1812707352),
DecisionTreeClassifier(max_features='sqrt', random_state=1822420100),
DecisionTreeClassifier(max_features='sqrt', random_state=1387906012),
DecisionTreeClassifier(max_features='sqrt', random_state=1410458496),
DecisionTreeClassifier(max_features='sqrt', random_state=1170539670),
DecisionTreeClassifier(max_features='sqrt', random_state=2033171363)]

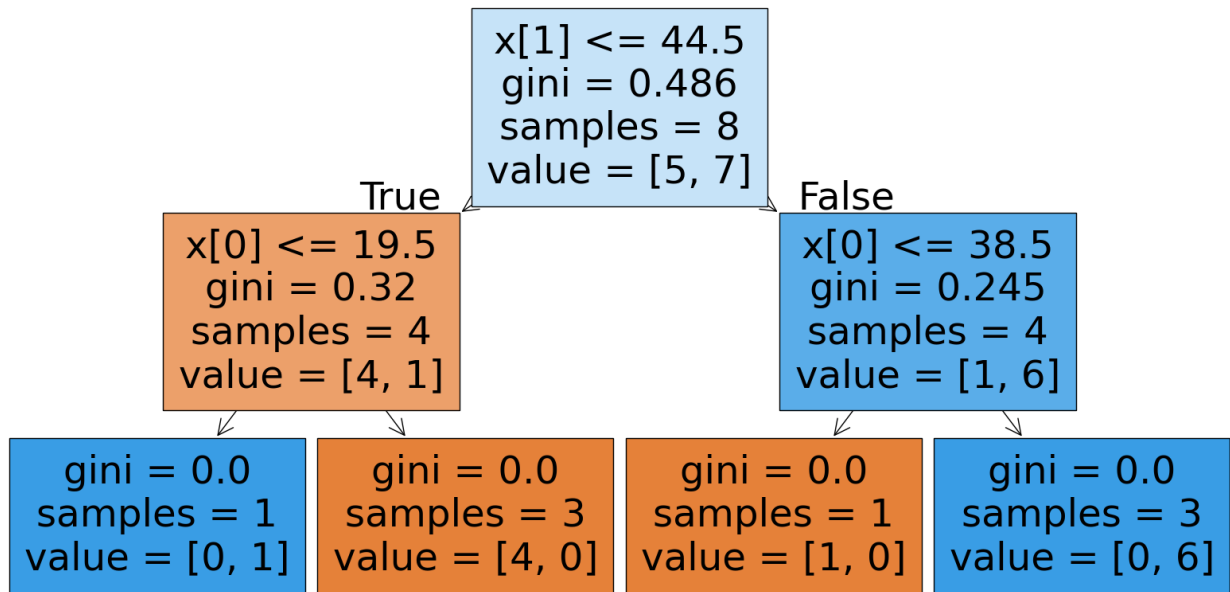
estimator=RF.estimators_[0]

from matplotlib import pyplot as plt
from sklearn.tree import plot_tree

plt.figure(figsize=(20,10))
plot_tree(estimator,filled=True)

[Text(0.5, 0.8333333333333334, 'x[1] <= 44.5\ngini = 0.486\nsamples =
8\nvalue = [5, 7]'),
Text(0.25, 0.5, 'x[0] <= 19.5\ngini = 0.32\nsamples = 4\nvalue = [4,
1]'),
Text(0.375, 0.6666666666666667, 'True '),
Text(0.125, 0.16666666666666666, 'gini = 0.0\nsamples = 1\nvalue =
[0, 1]'),
Text(0.375, 0.16666666666666666, 'gini = 0.0\nsamples = 3\nvalue =
[4, 0]'),
Text(0.75, 0.5, 'x[0] <= 38.5\ngini = 0.245\nsamples = 4\nvalue = [1,
6]'),
Text(0.625, 0.6666666666666667, ' False'),
Text(0.625, 0.16666666666666666, 'gini = 0.0\nsamples = 1\nvalue =
[1, 0]'),
Text(0.875, 0.16666666666666666, 'gini = 0.0\nsamples = 3\nvalue =
[0, 6]')]

```

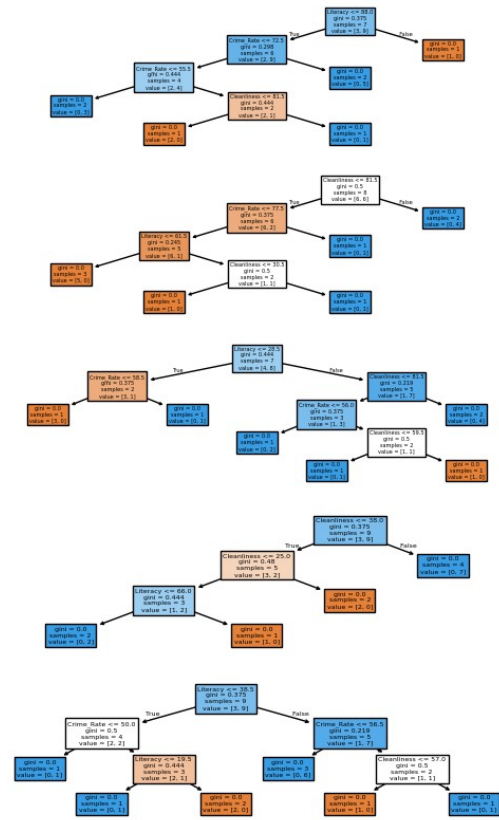
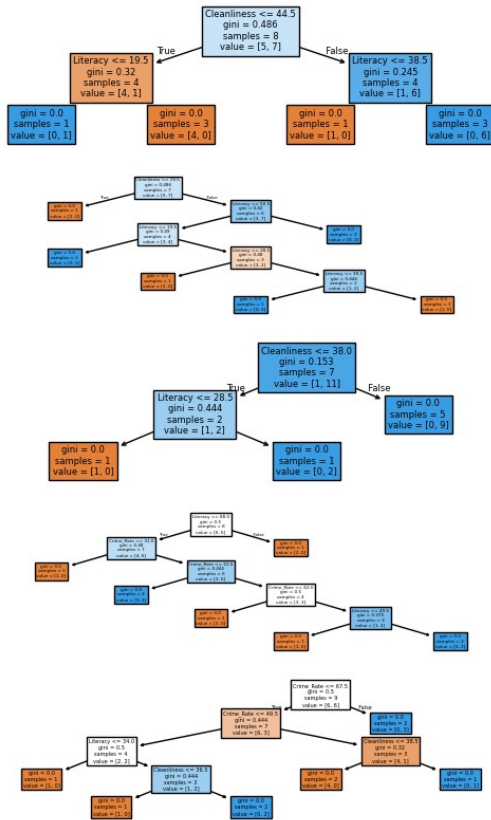


```

fig=plt.figure(figsize=(15,10))
for i in range(5):
    for j in range(2):
        fig.add_subplot(5,2,i*2+j+1)

plot_tree(RF.estimators_[i*2+j],feature_names=['Literacy','Cleanliness',
', 'Crime_Rate'], filled=True)

```



```
from matplotlib import pyplot as plt
for i in range(0,10):

plot_tree(RF.estimators_[i],feature_names=['Literacy','Cleanliness','Crime_Rate'],filled=True)
plt.show()
```

